Machine learning
Intro and basics

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Erasmus class – Tutoring slot: Thursdays, 11am – 1pm

(Not Machine Learning stuff)

National emergency numbers

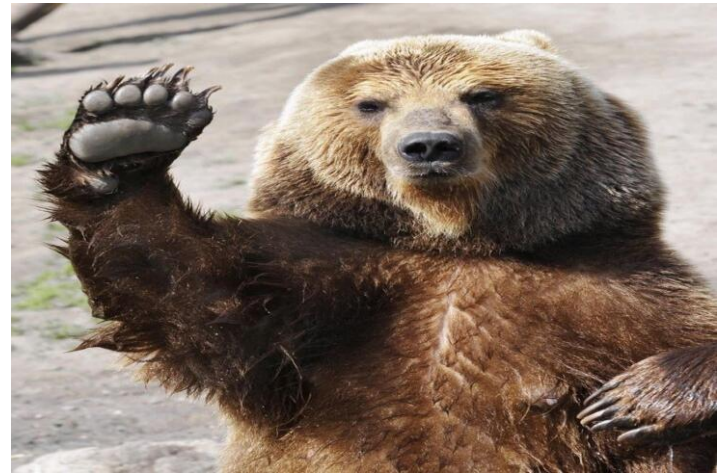
100 - Hellenic Police Number
1056 - National Child Protection Helpline
108 - Hellenic Coast Guard Number
112 - European emergeny service line
116000 - European Hotline for Missing Children
116111 - EU Helpline for Children and Teenagers
166 - Medical Emergency Service (EKAB) Number
199 - Hellenic Fire Service Number

Websites with information on local events, news etc (not affiliated with the Uni). They have social media accounts too. Sadly, they are in greek.

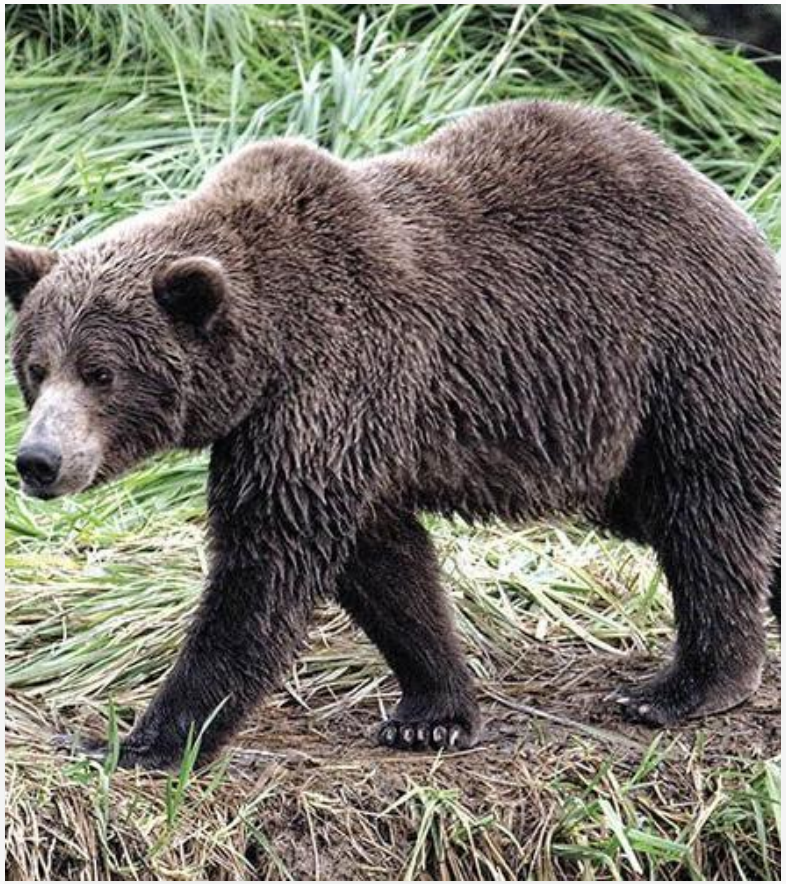
- **https://www.sindetiras.gr/**
- **https://www.corfuland.gr/**

We have a Study group with some other students, holding journal clubs, chatting about Informatics and science in general (in English). We have a special interest on the crossroad between Computer Science and health metrics! If interested, let me know and I can add you to the group in Teams!

Machine learning in a nutshell



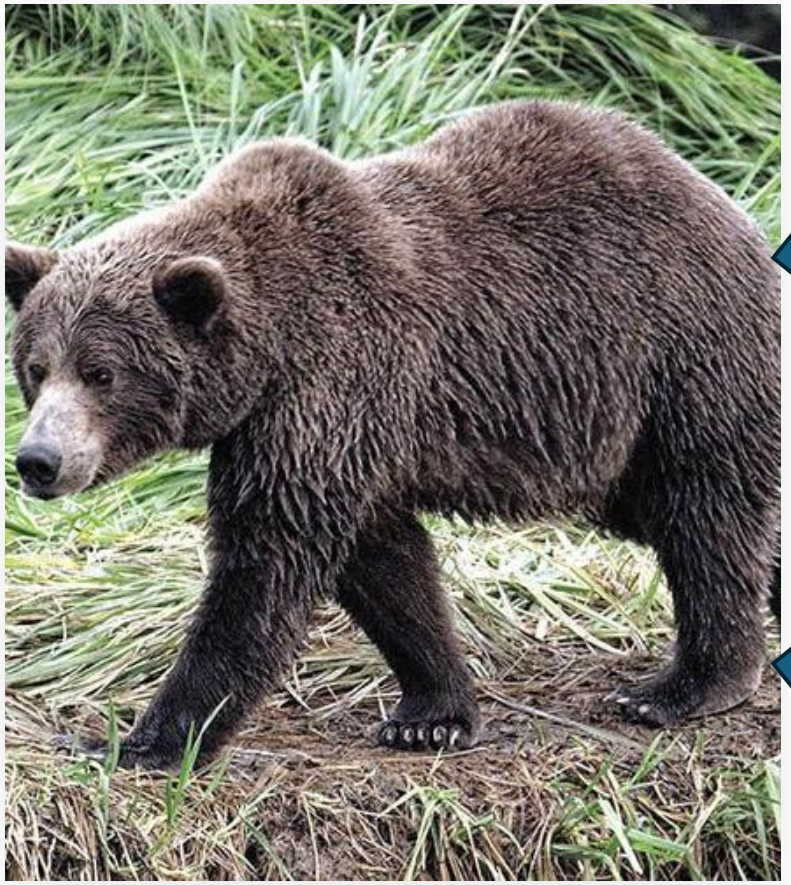
- Features of a bear:
  - Big size
  - Big muscles
  - Brown color
  - Fur
  - Big jaws
  - Lives in the forests
  - Omnivore
- Hypothesis: If it doesn't have **all** of the above features, then it's not a bear







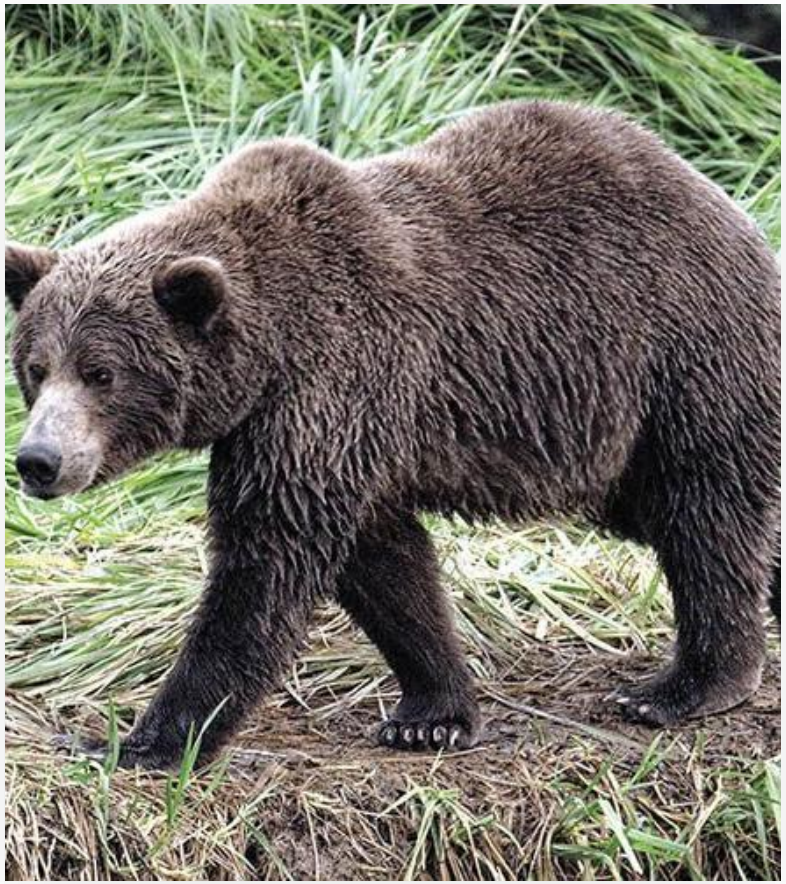




















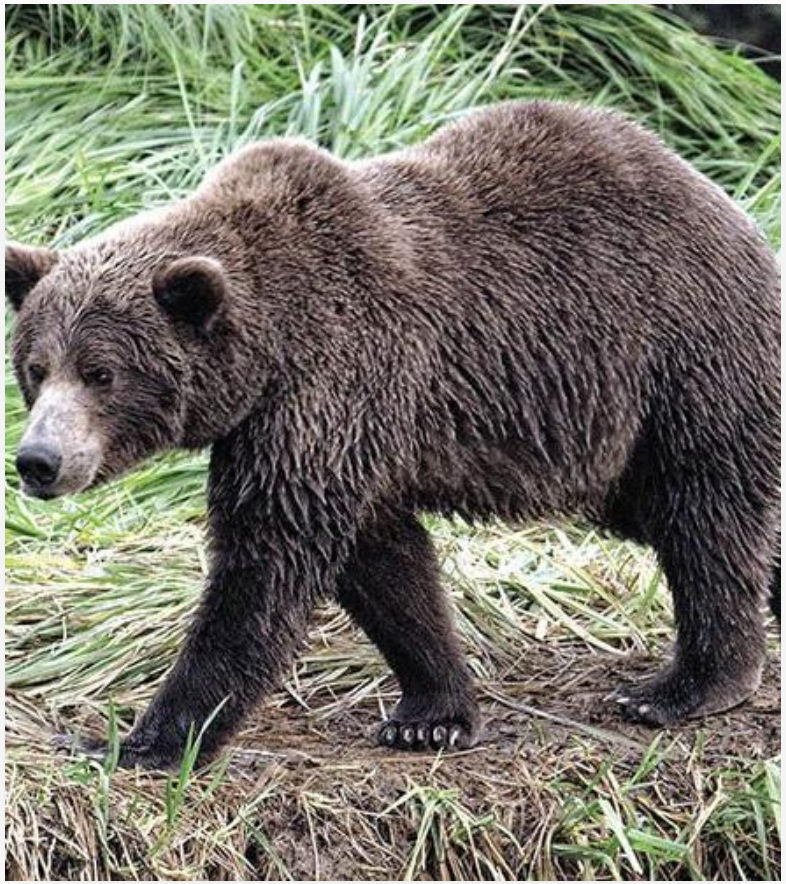






Machine learning in a nutshell

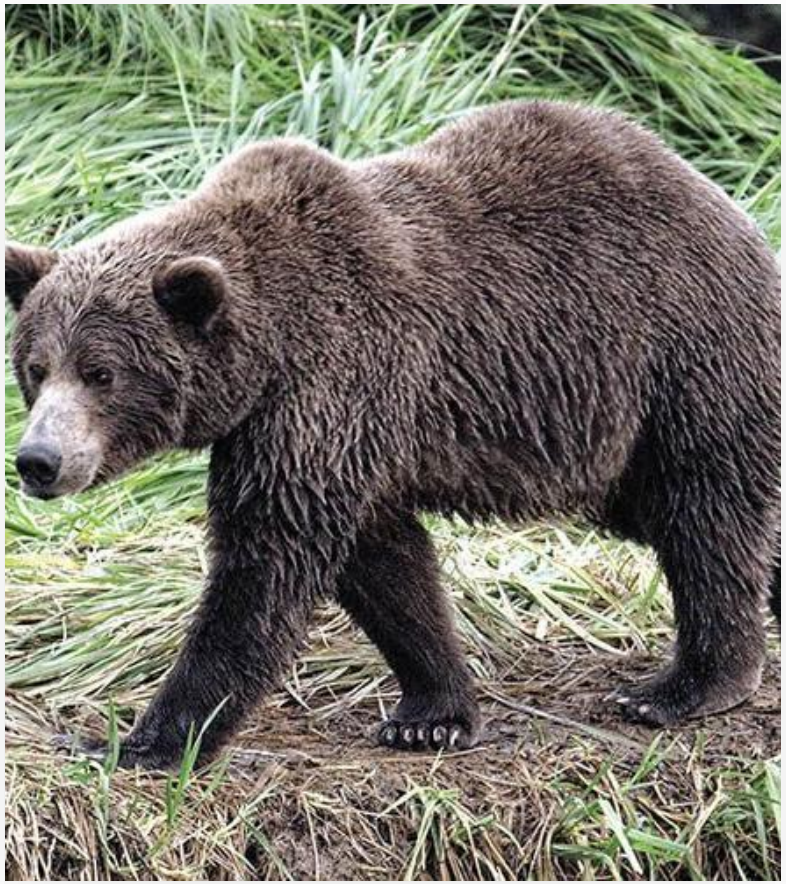
- Features of a bear:
  - Big size
  - Big muscles
  - ~~Brown color~~ Brown OR white color
  - Fur
  - Big jaws
  - ~~Lives in the forests~~ Lives in the forests OR on the ice
  - ~~Omnivore~~ Omnivore OR carnivore
- Hypothesis: If it doesn't have **all** of the above features, then it's not a bear

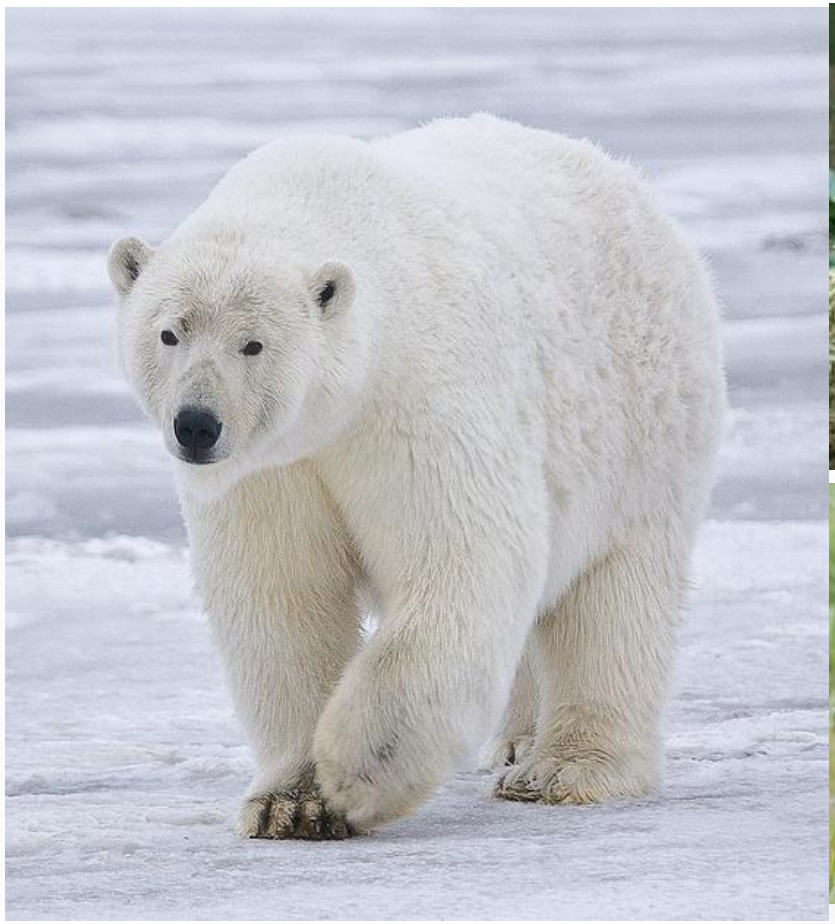






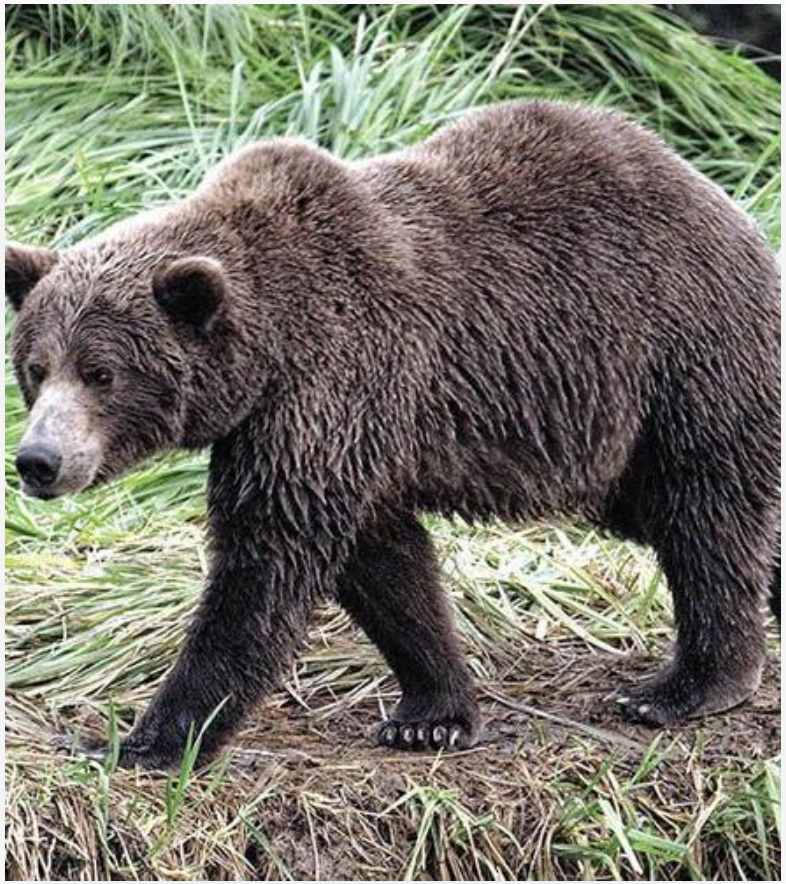


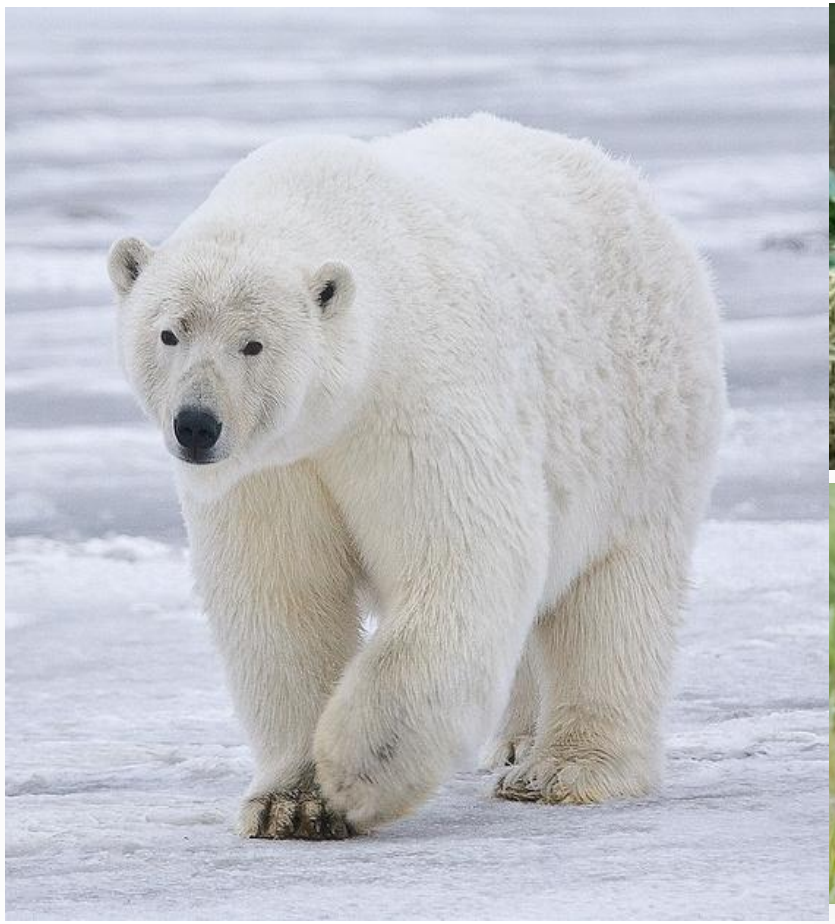
















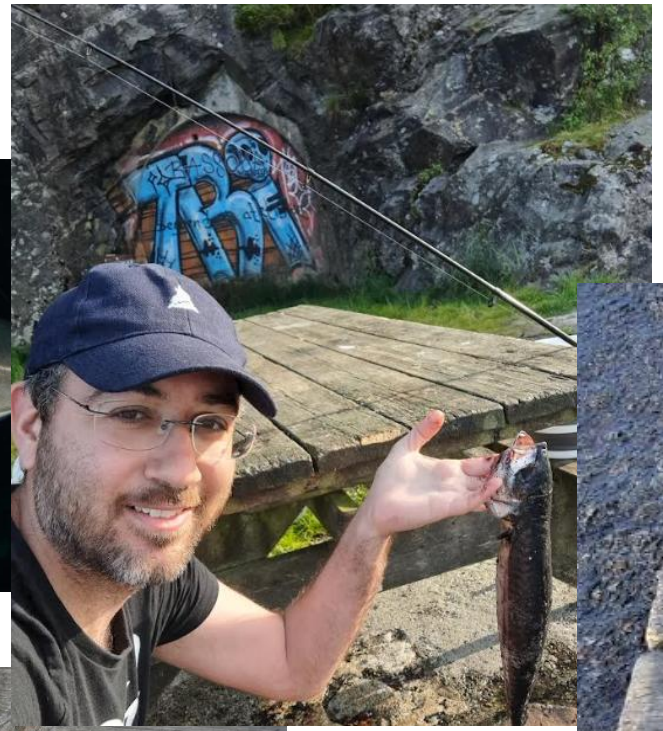
Machine learning in a nutshell

- Features of a bear:
  - Big size
  - Big muscles
  - ~~Brown color~~ Brown OR white color ~~Brown OR white color~~ Brown OR white color OR black/white
  - Fur
  - Big jaws
  - ~~Lives in the forests~~ Lives in the forests OR on the ice
  - ~~Omnivore Omnivore OR carnivore~~ Omnivore OR carnivore OR hebivore
- Hypothesis: If it doesn't have **all** of the above features, then it's not a bear

It’s not always that obvious!











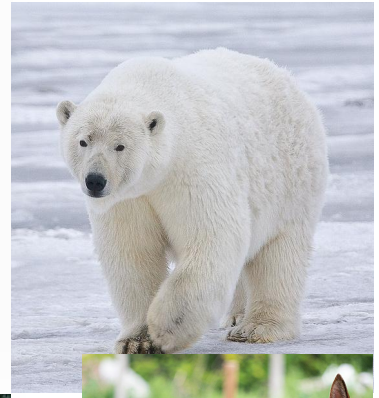


Training

Inference

Testing



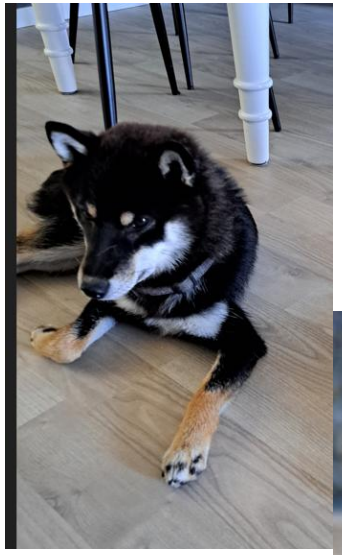


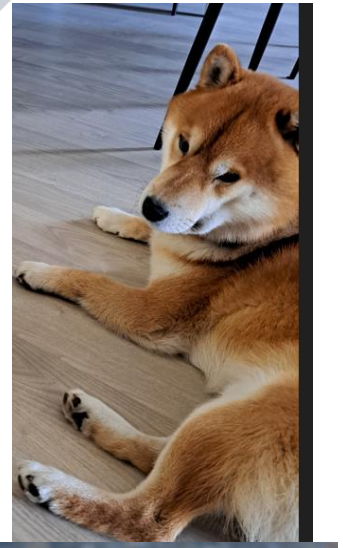






- Big size
- Big muscles
- Brown, white or black/white
- Fur
- Big jaws
- Live in the forests or on ice
- Omnivorous or carnivorous or herbivorous





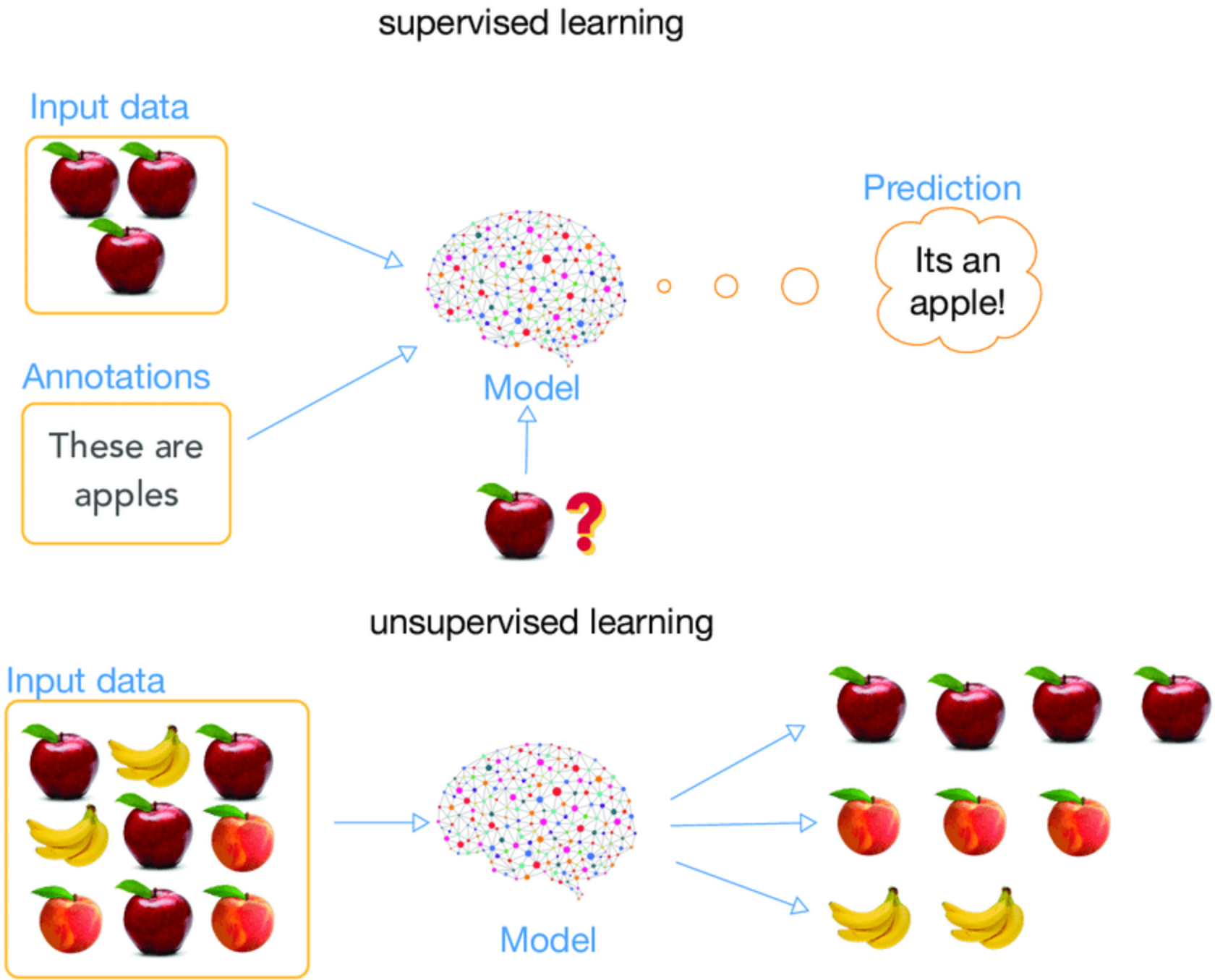


Main types of ML

- Supervised (classification)
  - Decision Trees
  - k-nearest neighbors (aka kNN)
  - Random forests (optional)

- Unsupervised (clustering)
  - Kmeans
  - Hierarchical clustering
  - DBSCAN (optional)




supervised learning
Input data
Annotations
These are apples
Model
Prediction
Its an apple!
unsupervised learning
Input data
Model

Study at your own pace

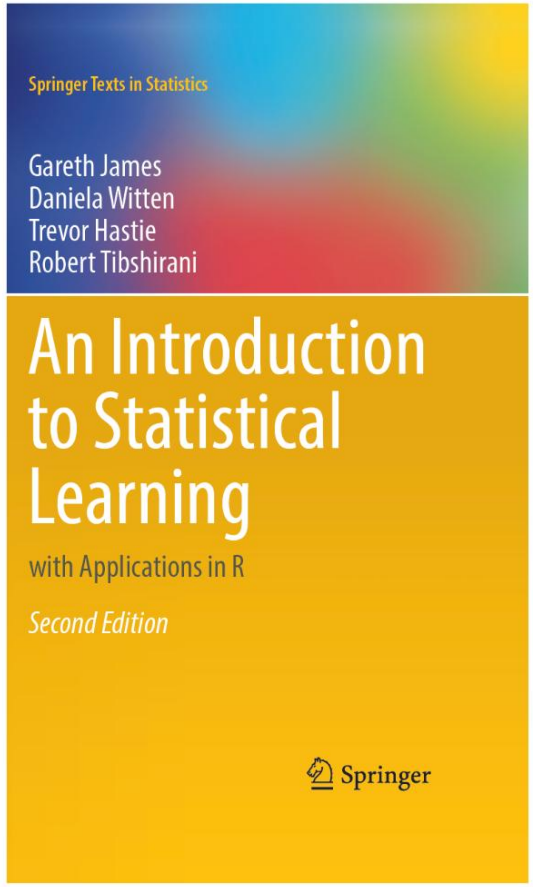
- Suggested order of study, check the Word document ->study_guide.docx
  - Will be updated as we make progress through the semester!

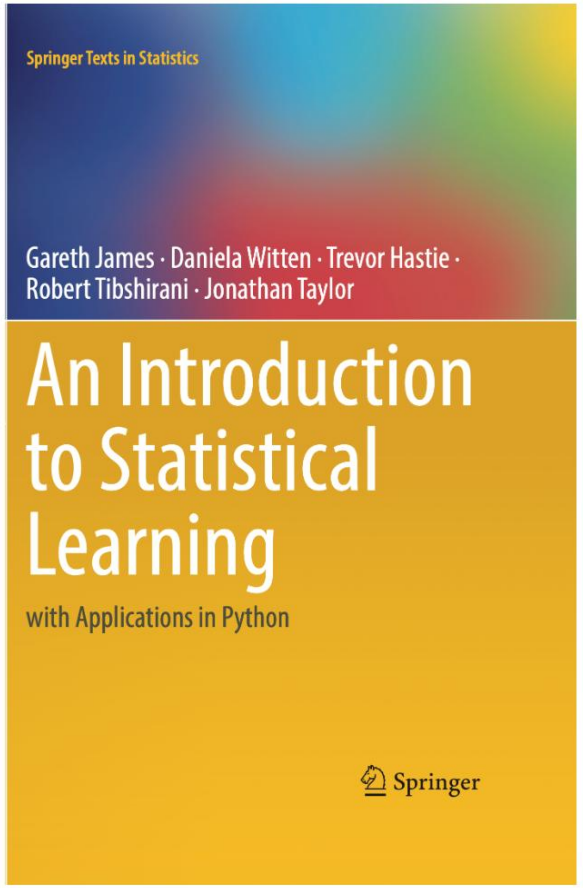
- Some dates are still TBD, I'm waiting for your input!

- Useful additional resources to help you with ML and the course:
  - Hands-On Machine Learning with R (ignore that they use R, it's still a great resource to learn about ML)
  - Suggested textbooks (all free) in the next 2 slides

Useful resources

- Available for free at https://www.statlearning.com/





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Also useful if curious about statistics (not necessary for our course)

- https://link.springer.com/book/10.1007/978-0-387-21736-9
  - Available for free at https://www.stat.cmu.edu/~brian/valerie/617-2022/0%20-%20books/2004%20-%20wasserman%20-%20all%20of%20statistics.pdf

We plan to have two midterms (graded as **bonus**)

- One on supervised learning
  - Date: TBD
- One on unsupervised learning
  - Date: TBD

Final project (detailed instructions coming soon)

Apply the algorithms to a dataset (you can use *Python* or *R*)

1. Apply basic preprocessing (for example, standardize numerical values) and exploration (for example, plot histograms, correlations etc)
2. Apply the algorithms we learnt
3. Submit a short report
4. Present the findings (10 min each, followed by discussion)

Keywords to take home

- Uncertainty
- Visualization
- Statistical tests
- Verification
- Update the model
- Bias and bias correction

Why ML: Massive (Big) Data

- Timeseries
- Business analytics
- Biology
- Public health

- Cross-tabulated? Text? Multimedia? Multi-modal?
- Raw? Processed?
- Pruned?
- Missing observations?
- Data imputation?

- Synthetic? Real?
- Sensitive?
- Outliers?

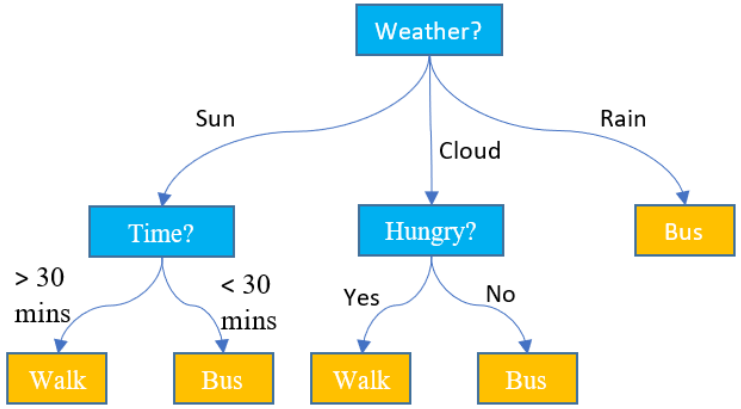
Classification

- Binary vs multi-class classification

- In a classification task, we are supposed to predict discrete target variables(class labels) using independent features.
- In the classification task, we are supposed to find a decision boundary that can separate the different classes in the target variable.

- Types of Classification Algorithms
- There are different types of State of the art classification algorithms that have been developed over time to give the best results for classification tasks by employing techniques like bagging and boosting.

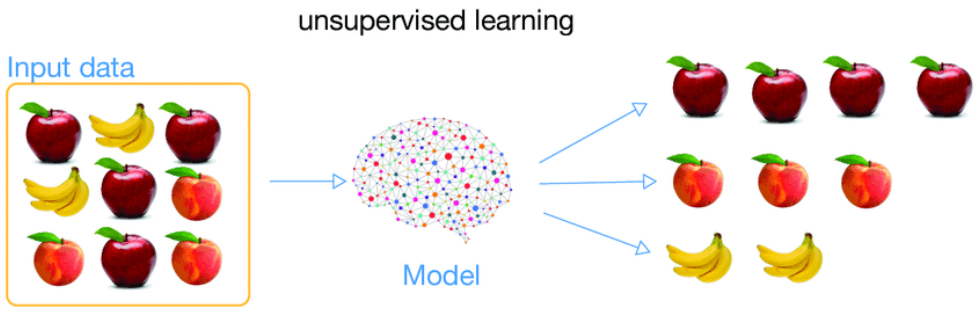
- Decision Tree
- Random Forest Classifier
- K – Nearest Neighbors
- Support Vector Machine (not part of the curriculum)



Clustering

- Unsupervised learning

- We see them later in the semester



Regression (just a refresher, not part of the curriculum)

- In a regression task, we are supposed to predict a continuous target variable using independent features.
- In the regression tasks, we are faced with generally two types of problems linear and non-linear regression.

- Types of Regression Algorithms
- There are different types of State of the art regression algorithms that have been developed over time to give the best results for regression tasks by employing techniques like bagging and boosting.

- Lasso Regression
- Ridge Regression
- XGBoost Regressor
- LGBM Regressor

Terminology

- **variable**, also known as **attribute** or **feature** or **predictor** or **independent variables**. If you have patients in rows, variables are in columns and they describe these patients (i.e. with name, date of birth, heart rate, and so on).
- **data sample**, also known as **data instance**. These are the rows in your table and represent the objects or subjects you are researching. For example, patients, transactions, etc
- **categorical** or **discrete** variable. This refers to a variable with categories as values, for example eye color, nationality, clothe type, and so on.
- **numeric** or **continuous** variable. This refers to a variable with numbers as values, for example height, age, and so on.
- **string** or **text** variable. This refers to a variable with text as value, for example the address, a person's name etc.

Some stuff on using R – Skip if planning to use Python

Base R
Cheat Sheet

Getting Help

Accessing the help files

?mean
Get help of a particular function.
help.search('weighted mean')
Search the help files for a word or phrase.
help(package = 'dplyr')
Find help for a package.

More about an object

str(iris)
Get a summary of an object's structure.
class(iris)
Find the class an object belongs to.

Using Libraries

install.packages('dplyr')
Download and install a package from CRAN.

library(dplyr)
Load the package into the session, making all its functions available to use.

dplyr::select
Use a particular function from a package.

data(iris)
Load a built-in dataset into the environment.

Working Directory

getwd()
Find the current working directory (where inputs are found and outputs are sent).

setwd('C://file/path')
Change the current working directory.

Use projects in RStudio to set the working directory to the folder you are working in.

Vectors

Creating Vectors

c(2, 4, 6)	2 4 6	Join elements into a vector
2:6	2 3 4 5 6	An integer sequence
seq(2, 3, by=0.5)	2.0 2.5 3.0	A complex sequence
rep(1:2, times=3)	1 2 1 2 1 2	Repeat a vector
rep(1:2, each=3)	1 1 1 2 2 2	Repeat elements of a vector

Vector Functions

sort(x)
Return x sorted.
rev(x)
Return x reversed.
table(x)
See counts of values.
unique(x)
See unique values.

Selecting Vector Elements

By Position

x[4]	The fourth element.
x[-4]	All but the fourth.
x[2:4]	Elements two to four.
x[-(2:4)]	All elements except two to four.
x[c(1, 5)]	Elements one and five.

By Value

x[x == 10]	Elements which are equal to 10.
x[x < 0]	All elements less than zero.
x[x %in% c(1, 2, 5)]	Elements in the set 1, 2, 5.

Named Vectors

x['apple']	Element with name 'apple'.

Programming

For Loop

```
for (variable in sequence){
   Do something
}
```

Example

```
for (i in 1:4){
   j <- i + 10
   print(j)
}
```

While Loop

```
while (condition){
   Do something
}
```

Example

```
while (i < 5){
   print(i)
   i <- i + 1
}
```

If Statements

```
if (condition){
   Do something
} else {
   Do something different
}
```

Example

```
if (i > 3){
   print('Yes')
} else {
   print('No')
}
```

Functions

```
function_name <- function(var){
   Do something
   return(new_variable)
}
```

Example

```
square <- function(x){
   squared <- x*x
   return(squared)
}
```

Reading and Writing Data

Input	Ouput	Description
df <- read.table('file.txt')	write.table(df, 'file.txt')	Read and write a delimited text file.
df <- read.csv('file.csv')	write.csv(df, 'file.csv')	Read and write a comma separated value file. This is a special case of read.table/ write.table.
load('file.RData')	save(df, file = 'file.Rdata')	Read and write an R data file, a file type special for R.

Conditions

a == b	Are equal	a > b	Greater than	a >= b	Greater than or equal to	is.na(a)	Is missing
a != b	Not equal	a < b	Less than	a <= b	Less than or equal to	is.null(a)	Is null

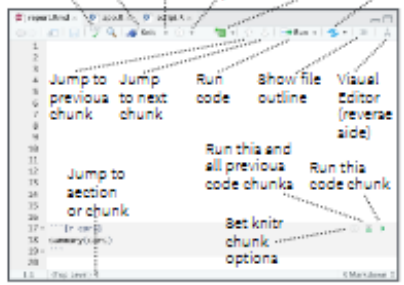
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Google *'Base R Cheat Sheet'*

RStudio IDE : : CHEAT SHEET

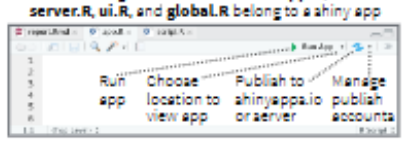
Documents and Apps

Open Shiny, R Markdown, knitr, Sweave, LaTeX, .Rd files and more in Source Pane

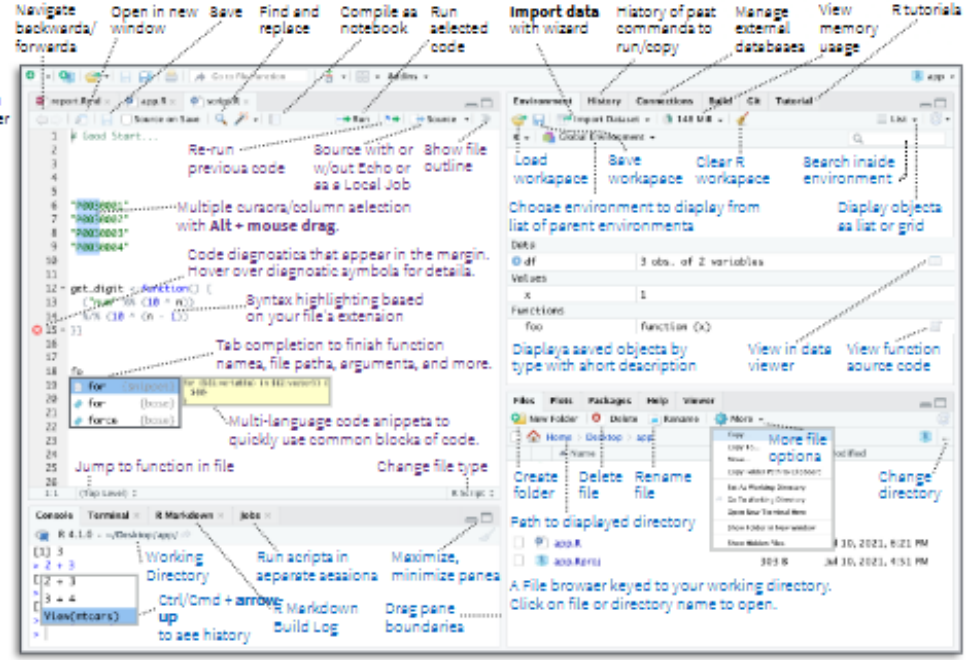


Access markdown guide at **Help > Markdown Quick Reference**
See reverse side for more on **Visual Editor**

RStudio recognizes that files named **app.R**, **server.R**, **ui.R**, and **global.R** belong to a shiny app



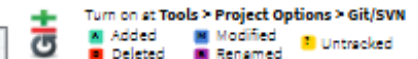
Source Editor

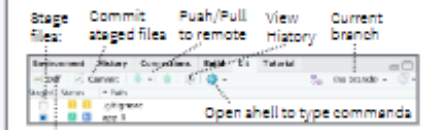


Tab Panes

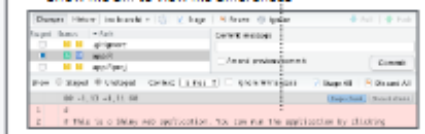
Version Control

Turn on at **Tools > Project Options > Git/SVN**



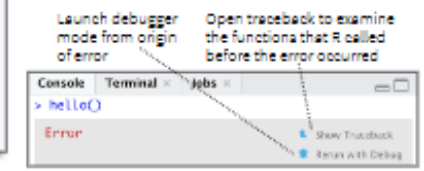


Show file diff to view file differences



Debug Mode

Use **debug()**, **browser()**, or a breakpoint and execute your code to open the debugger mode.



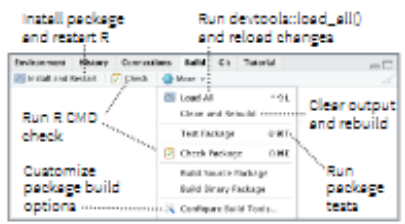
Package Development

Create a new package with **File > New Project > New Directory > R Package**

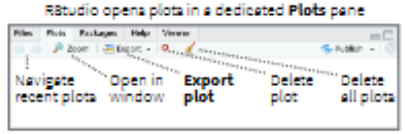
Enable roxygen documentation with **Tools > Project Options > Build Tools**

Roxygen guide at **Help > Roxygen Quick Reference**

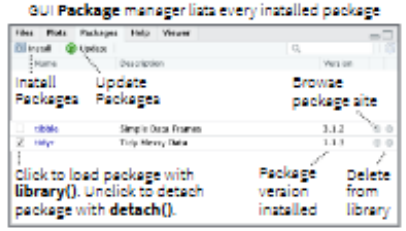
See package information in the **Build Tab**



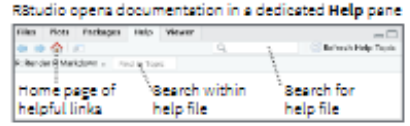
RStudio opens plots in a dedicated **Plots** pane



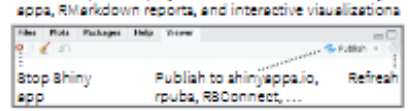
GUI **Package** manager lists every installed package



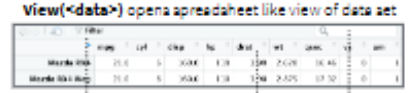
RStudio opens documentation in a dedicated **Help** pane



Viewer pane displays HTML content, such as Shiny apps, RMarkdown reports, and interactive visualizations



View(<data>) opens spreadsheet like view of data set



Click next to line number to add/remove a breakpoint.

Highlighted line shows where execution has paused



Run commands in environment where execution has paused

Examine variables in executing environment

Select function in traceback to debug



Google *'Base R Cheat Sheet'*

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Start a new script in Rstudio

```
# working directory
getwd()
setwd("C:/Users/kgi062/ML_2025")
print("Hello, world!")
2 + 2
x <- 10         # Numeric
name <- "Alice"   # Character
flag <- TRUE     # Logical
numbers <- c(1, 2, 3, 4, 5)
mean(numbers)
sum(numbers)
?CO2
View(CO2)
head(CO2)
tail(CO2)
colnames(CO2)
dim(CO2)
str(CO2)
summary(CO2)
summary(CO2$uptake)
median(CO2$uptake)
class(CO2$uptake)
unique(CO2$Treatment)
subset(CO2, conc == min(CO2$conc))
hist(CO2$uptake)
plot(Orange$age, Orange$circumference)
plot(Orange$age, Orange$circumference,
   xlab="Age", ylab="Circumference",
   main="Circumference vs. Age",
   col="blue", pch=16)

# data come from an unknown normal distribution
data <- read.csv("external_file.csv")
d = rnorm(100,mean = 5,sd=1)
plot(data)
plot(d)
```

Copy and paste the code to run it in R/Rstudio